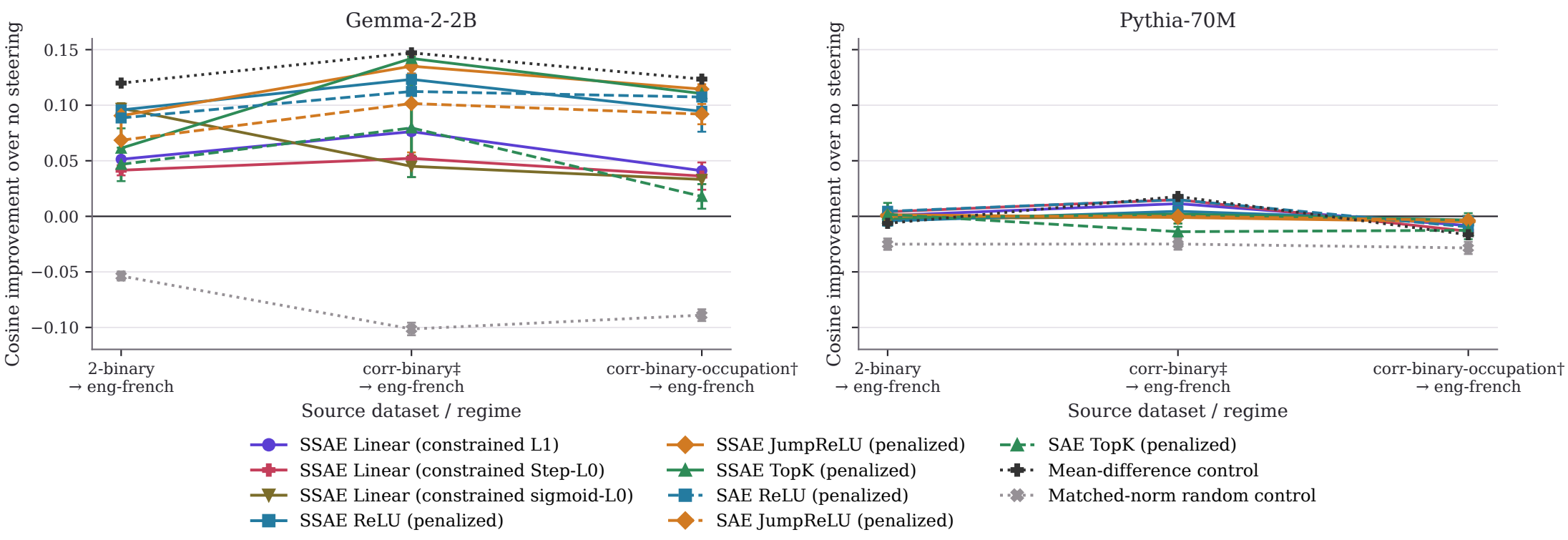
OOD steering improvement
Gemma-2-2B
Pythia-70M
Cosine improvement over no steering
0.15
0.10
0.05
0.00
−0.05
−0.10
2-binary
→ eng-french
corr-binary‡
→ eng-french
corr-binary-occupation†
→ eng-french
Source dataset / regime
SSAE Linear (constrained L1)
SSAE Linear (constrained Step-L0)
SSAE Linear (constrained sigmoid-L0)
SSAE ReLU (penalized)
SSAE JumpReLU (penalized)
SSAE TopK (penalized)
SAE ReLU (penalized)
SAE JumpReLU (penalized)
SAE TopK (penalized)
Mean-difference control
Matched-norm random control